

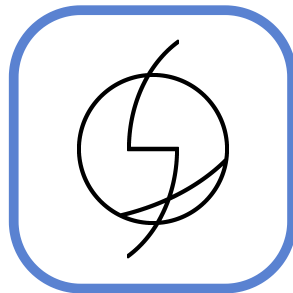
A course in language creation



Conlangs University

Phonology — Class 1

The IPA, phonemes, and basic notation



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Prerequisites

Please watch this [video by youtuber Tom Scott](#). It is a very concise introduction to the International Phonetic Alphabet.

You can find an interactive (clickable) chart in [this webpage](#) from the University of Victoria, Canada. [The TypeIt app for IPA](#) is a simple way to type the symbols you might not be able to access easily from your regular keyboard, though [other ways exist](#).

The video above will be completed by an extensive section in this document explaining the IPA and how to read and use it.

The study of sounds

The study of speech sounds is called [phonetics](#). It is the basis for phonological analysis.

[Phonology](#) is the study of the patterns sounds form in language.

The two disciplines are very often confused, due to their tight relationship and the (partial) reliance of one on the other. Here's a summary of the differences:

Phonetics	Phonology
Is the basis for phonological analysis.	Is the basis for further work in morphology, syntax, discourse, and orthography design.
Analyses the production and acoustics of all human speech sounds, regardless of language.	Analyses the sound patterns of a particular language by determining which phonetic sounds are significant, and explaining how these sounds are interpreted by the native speaker.

Basic speech

To produce a sound, you (generally) need to expell air from your lungs, creating a steady flow through your windpipe. With your speech organs, you can affect that sound.

- The vocal cords can vibrate or not (voicing);
- The shape of the mouth and other speech organs affects the resonance of the vocal tract, and thus the resulting sound;
- The nasal cavity adds resonance for some sounds (see [nasal consonants](#) for instance);

The IPA

In order to express sound through text, instead of using English equivalents that more often than not miss the mark and fail to accurately convey the sounds, we use a dedicated system: the [International Phonetic Alphabet, or IPA](#).

The IPA is intended to be a universal (well, international) system of transcription, able to convey the sounds of any and all human spoken languages, although it is heavily based on the Latin alphabet. The reason for this reliance on this particular writing system is that it is the most used worldwide, by the most languages and speakers.

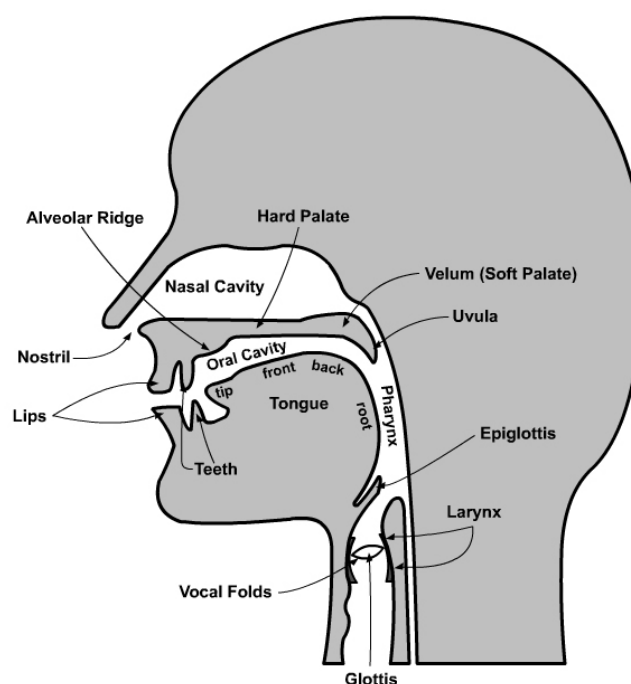
Throughout this document, we will be making references to [the interactive chart provided by the University of Victoria](#).

Consonants

The IPA's consonants chart is based on two axes:

Place of articulation (left-right axis), which is where in the mouth two *articulators* (speech organs) are interacting, and **manner of articulation** (top-down axis), or how the airflow is controlled.

To offer you some points of reference for how *place and manner of articulation* work and interplay, here are a few images. You may have seen the third one floating around online.



This one is a schematic representation of the different *places of articulation*, indicating where in your vocal tract (the ensemble of the organs involved in physical speech production) sounds are produced.

To produce a sound, two articulators need to interact. Very often it is the tongue, as the most mobile articulator, that goes to meet another part of the mouth.

Those articulators can be *passive* or *active*. A *passive* articulator is one that does not move, while an *active* articulator is one that moves towards the passive one.

Often, in names that include two articulators, such as "linguolabial", the *active articulator* is the first mentioned (here, the tongue) and the *passive articulator* second (here, the lip).

If only one is mentioned, generally the tongue is the *active articulator*. For instance, "dental" means the tongue moves (active articulator) to touch the teeth (passive articulator). You could try the opposite, but good luck moving your teeth with all your non-existent teeth muscles.

Some articulations explicitly rule out the tongue as an articulator: "labiodental" for instance, means that the bottom lip makes contact with the upper teeth, and "bilabial" means both lips are involved.

Other articulations implicitly rule the tongue out: you can't touch your glottis with your tongue... Right? That would be very weird.

CONSONANTS (PULMONIC)

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
<u>Plosive</u>	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
<u>Nasal</u>	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
<u>Trill</u>	ʙ			r					ʀ		
<u>Tap or Flap</u>		ɸ		ɾ		ɽ					
<u>Fricative</u>	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	ħ ʕ	ʕ ʕ	h ɦ
<u>Lateral fricative</u>				ɬ ɮ							
<u>Approximant</u>		ʋ		ɹ		ɻ	j	ɰ			
<u>Lateral approximant</u>				l		ɭ	ʎ	ʟ			

Where symbols appear in pairs, the one to the right represents a voiced consonant. Shaded areas denote articulations judged impossible.

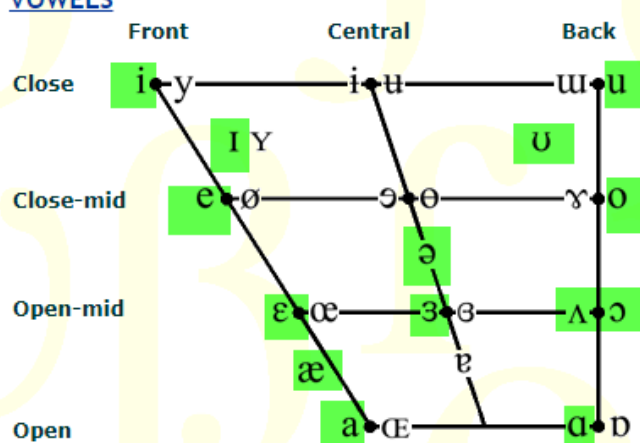
CONSONANTS (NON-PULMONIC)

Clicks	Voiced implosives	Ejectives
◌ Bilabial	ɓ Bilabial	ʼ Examples:
Dental	ɗ Dental/alveolar	pʼ Bilabial
! (Post)alveolar	ɟ Palatal	tʼ Dental/alveolar
‡ Palatoalveolar	ɡ Velar	kʼ Velar
Alveolar lateral	ɠ Uvular	sʼ Alveolar fricative

OTHER SYMBOLS

ɱ Voiceless labial-velar fricative	ɕ ʑ Alveolo-palatal fricatives
ɰ Voiced labial-velar approximant	ɺ Alveolar lateral flap

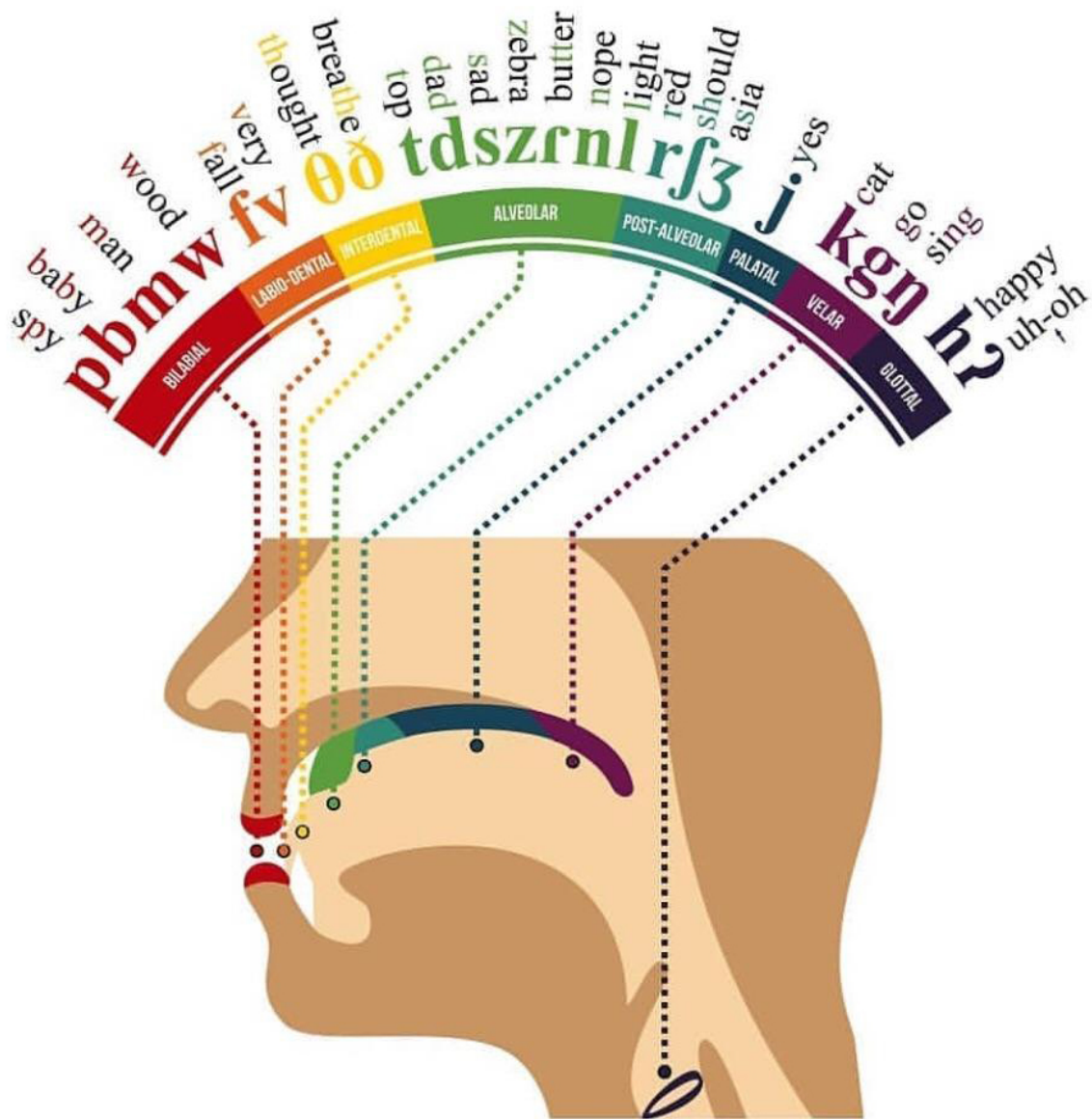
VOWELS



Where symbols appear in pairs, the one to the right represents a rounded vowel.

Here is a simplified inventory of the sounds of English. The vowels are not particularly accurate as they do not display *diphthongs*, or sounds that consist of a vowel and a *glide* (or *semivowel*) but are analysed as a single sound unit when it comes to altering the meaning of a word (more on this in the *phoneme* part of this article).

The consonants are lacking two sounds not displayed in the chart: the *affricates* ʈʂ and ɖʐ (which you can hear at the end of "touch" and the beginning of "jump", respectively). Affricates are (roughly, for now) sound units starting with a *stop* but ending in a *fricative*.



This one is a rough representation of most of the phonemic consonants of English. It also contains **ɹ**, a sound present in some dialects (chiefly in American English) as an alternative to **r** in certain phonological contexts.

The **ɹ** sound is represented with **r**, as it is easier to type and recognise for English speakers, whom this chart is aimed at.

In addition to the two axes we've talked about, the consonant chart divides each of its cells into two sides: on the left, the unvoiced sounds, and on the right, the voiced sounds. We call that difference *voicing*.

Those two variants of a sound are produced in the same place in your mouth, and the only difference is whether or not your vocal cords are vibrating while producing it.

For instance, try saying "fail" (feɪl) and "veil" (veɪl), or "thy" (ðaɪ) and "thigh" (θaɪ). Place your hand on your throat to feel it vibrate or not!

This difference can also be felt when comparing normal speaking voice with whispering.

Places of articulation

Here, we distinguish the tongue *tip* (the very end of the tongue) from the tongue *blade* (the area just behind the tip).

- **Bilabial:** both lips
- **Labiodental:** top teeth and lower lip
- **Dental:** tongue between the teeth or against the inner face of the upper teeth
- **Alveolar:** tip of the tongue against the alveolar ridge (the little bump behind the top teeth)
- **Postalveolar:** tongue blade just behind the alveolar ridge
- **Palatal:** tongue blade further behind the ridge, around where the roof of the mouth is highest
- **Retroflex:** tip of the tongue slightly curled back
- **Velar:** tongue against the velum (soft palate)
- **Uvular:** tongue against the uvula (the punchingbag-looking sack of meat dangling at the end of your mouth)
- **Pharyngeal:** tongue against the pharynx
- **Glottal:** articulation involving the *glottis* (the aperture between the vocal cords)

Manners of articulation

- **Stop/Plosive:** closure of the airflow, then release
- **Nasal:** closure of the airflow in the mouth, with a velic (see *velar*) opening allowing the air through the *nasal cavity*
- **Trill:** rapid closures and openings
- **Tap/Flap:** one rapid closure and opening
- **Fricative:** narrowing of the vocal tract resulting in a turbulent airflow
- **Lateral fricative:** same as fricative, with the air escaping to the sides of the tongue
- **Approximant:** a narrowing of the vocal tract, but not enough to cause a turbulent airflow
- **Lateral approximant:** same as approximant, but with the air escaping to the sides of the tongue

Articulators

Active articulators

Those that move.

- Lips (mostly the lower one)
- Tongue
- Uvula
- Glottis

Passive articulators

Those that don't move on their own.

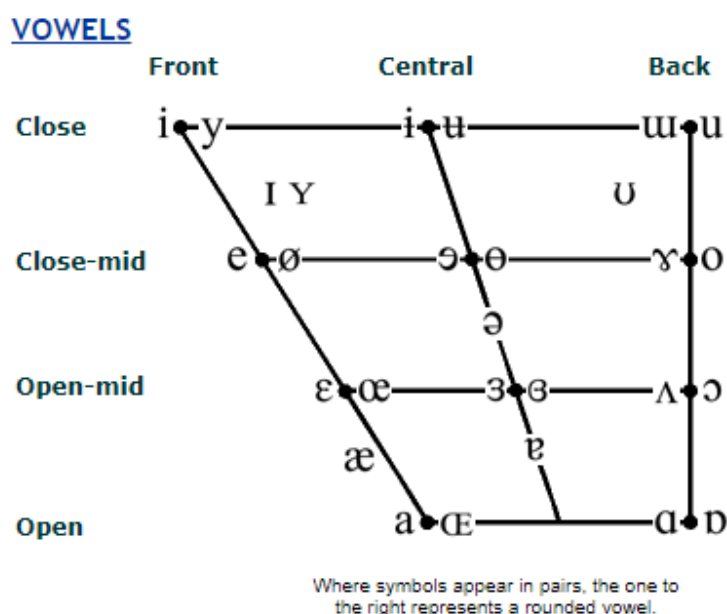
- Teeth (upper and lower)
- Alveolar ridge
- Palate
- Velum

The pharynx is a contact area for the production of some consonants (pharyngeal consonants).

The nasal cavity is used as a resonance chamber for the production of the nasal consonants.

Vowels

I encourage you to read the entire [wikipedia article about vowels](#), as it contains information that completes what is given here.



The vowels chart is also laid out to represent how and where the sound is produced in the mouth.

The open-close axis is also sometimes labelled low-high, or *vowel height*, because it represents the position of the tongue and how high it is. The higher it is, the more it closes the vocal tract. As such, [i] is produced with a more closed vocal tract than [a]. Try it!

The front-back axis represent *vowel backness*, or how far back in the mouth the vowel is produced. Compare the vowels [i] and [u]: the main difference between them is the tongue being further back for [u], but its height does not change significantly.

Pairs of vowels around a same point on the diagram occur when *roundedness* is a distinctive feature (more on that later) of the vowel. In these cases, the left vowel is *unrounded* while the right one is *rounded*. Roundedness is when the lips form a rounder shape during sound production.

For instance, try pronouncing the "French u", which is noted as **y** on the chart. Start by saying [i], then round your lips without moving your tongue. Tada: you've produced [y], a sound notoriously difficult to consistently get for many learners of the French language.

Other symbols

Non-pulmonic consonants

CONSONANTS (NON-PULMONIC)

Clicks	Voiced implosives	Ejectives
◌ Bilabial	ɓ Bilabial	’ Examples:
◌ Dental	ɗ Dental/alveolar	p’ Bilabial
◌ (Post)alveolar	ɟ Palatal	t’ Dental/alveolar
◌ Palatoalveolar	ɠ Velar	k’ Velar
◌ Alveolar lateral	ɠ Uvular	s’ Alveolar fricative

These consonants are separated from the main chart because they do not rely on airflow from the lungs to produce sound.

Although none of them are present in English, you probably know some of the **clicks** as they are used as **paralinguistic signs** in many varieties of English. In English, you can express disapproval or pity by clicking your tongue right behind your teeth, which is often rendered as "tut-tut" or "tsk-tsk" in writing. It's also how many call a horse.

These sounds are said to be **paraphonemic**, as their usage is limited to specific contexts where their meaning can be understood easily, even though they're not part of the language itself.

Ejective consonants (**t’** **k’**) are characterised by a puff of air, additional to the pulmonic airflow and coming from the glottis. They are not *phonemic* in English, but ejective stops happen in some British dialects in specific contexts, notably at the end of word with a plosive, like "book".

Other consonants

OTHER SYMBOLS

ɱ	Voiceless labial-velar fricative	ç ʝ	Alveolo-palatal fricatives
w	Voiced labial-velar approximant	ɺ	Alveolar lateral flap
ɥ	Voiced labial-palatal approximant	ɧ	Simultaneous ʃ and x
ħ	Voiceless epiglottal fricative	Affricates and double articulations can be represented by two symbols joined by a tie bar if necessary.	
ʕ	Voiced epiglottal fricative		
ʡ	Epiglottal plosive		
		k͡p	t͡s

Those symbols are consonants that did not fit in the main chart because they either:

- make use of two places or manners of articulation (such as t͡ʃ or w)
- would take the space of a more common sound (for instance, ɺ would take the space of ɹ)

Their names are fairly self-explanatory if you understand the rest of the chart.

Diacritics

DIACRITICS

Diacritics may be placed above a symbol with a descender, e.g. $\underset{\cdot}{\eta}$

◦ Voiceless	$\underset{\cdot}{n}$ $\underset{\cdot}{d}$	.. Breathy voiced	$\underset{..}{b}$ $\underset{..}{a}$	̤ Dental	$\underset{̤}{t}$ $\underset{̤}{d}$
✓ Voiced	$\underset{✓}{s}$ $\underset{✓}{t}$	~ Creaky voiced	$\underset{~}{b}$ $\underset{~}{a}$	̣ Apical	$\underset{̣}{t}$ $\underset{̣}{d}$
^h Aspirated	t^h d^h	̤ Linguolabial	$\underset{̤}{t}$ $\underset{̤}{d}$	̤ Laminar	$\underset{̤}{t}$ $\underset{̤}{d}$
◌ More rounded	$\underset{◌}{ɔ}$	^w Labialized	t^w d^w	~ Nasalized	$\tilde{e}$
◌ Less rounded	$\underset{◌}{ɔ}$	^j Palatalized	t^j d^j	ⁿ Nasal release	d^n
⁺ Advanced	$\underset{+}{u}$	^Y Velarized	t^Y d^Y	^l Lateral release	d^l
◌ Retracted	$\underset{◌}{e}$	^ʕ Pharyngealized	$t^{ʕ}$ $d^{ʕ}$	◌ No audible release	$d^◌$
◌ Centralized	$\ddot{e}$	~ Velarized or Pharyngealized	$\underset{~}{t}$		
[×] Mid-centralized	$\overset{×}{e}$	◌ Raised	$\underset{◌}{e}$ ($\underset{◌}{I}$ =voiced alveolar fricative)		
◌ Syllabic	$\underset{◌}{n}$	◌ Lowered	$\underset{◌}{e}$ ($\underset{◌}{\beta}$ =lowered bilabial approximant)		
◌ Non-syllabic	$\underset{◌}{e}$	◌ Advanced Tongue Root	$\underset{◌}{e}$		
◌ Rhoticity	$\underset{◌}{ə}$ $\underset{◌}{a}$	◌ Retracted Tongue Root	$\underset{◌}{e}$		

The diacritics used by the IPA enable us to add phonetic detail to the "base" sounds.

Voiceless ($\underset{\cdot}{b}$ $\underset{\cdot}{m}$)

This one indicates that a sound is produced without vibrating the vocal cords. It can be added to a character representing a voiced sound to specifically indicate it's a variation of that sound, as opposed to a sound of its own, for example.

Voiced ($\underset{✓}{p}$ $\underset{✓}{f}$)

This one indicates a sound is produced with vibrating of the vocal cords.

More/less rounded ($\underset{◌}{ɔ}$ / $\underset{◌}{ɔ}$)

Those diacritics indicate the amount of *rounding* of the lips (which defines the *roundedness* of a vowel) involved in the production of the sound.

Advanced/Retracted ($\underset{+}{u}$ / $\underset{◌}{i}$)

Those diacritics indicate that the sound is pronounced further front or back, relatively to a predefined point of reference.

This phenomenon is called *fronting* and advanced/retracted can be expressed with the adjectives *fronted/backed* instead.

Centralised ($\ddot{a}$ $\ddot{e}$)

This diacritic serves to indicate that a vowel has undergone a shift towards the *central* part of the front-

back axis.

This phenomenon is called *centralisation*.

Mid-centralised (ɜ̌ ʌ̌)

This diacritic denotes a vowel having undergone a shift not only by way of *centralisation*, but also towards the mid- part of the open-close axis.

In other words, it has shifted closer to the mid-central vowel [ə].

Syllabic (ɹ̥ ɳ̥)

This diacritic applies to consonants, and resulting sounds are called *syllabic consonants* or *vocalic consonants*.

You can find them in English words like "rhythm" or "bottle".

In short, this means there is no vocalic content in a syllable but, as all syllables have to have a nucleus (more on that later), a consonant is taking this role.

Non-syllabic (ə̥ ʊ̥)

This diacritic is applied to vowels to signify that they are in fact semivowels, which are usually shorter and do not (phonemically or phonetically) behave the same way as normal vowels.

Rhoticity (ɐ̣ ɹ̣)

This hook diacritic symbolises an r-colored vowel, which is a vowel articulated along with a retroflex movement (tip/blade of the tongue turned up towards the palate), and sometimes with an additional constriction of the epiglottis.

Breathy voiced (ɹ̥ ɹ̥)

Also called *murmured voice*, this mode of voicing is attained by keeping the vocal cords further apart than usual, which increases the airflow in the larynx. You already know how to do it if you can whisper.

Creaky voiced (ə̰ ʒ̰)

Also called *vocal fry*, this **phonation** consists in compressing the vocal folds together without closing the airway completely.

Linguolabial (ɹ̥ ɹ̥)

Symbolises that the consonant is produced by placing the tip or blade of the tongue against the upper lip.

Aspirated (p^h b^h)

Aspiration is characterised by an additional/stronger puff of air when the consonant is released. Put a hand in front of your mouth and say "spin" ([spɪn]) then "pin" ([p^hɪn]). You should feel a difference!

Labialised (t^w k^w)

A labialised consonant is one whose production involves movement of the lips during, or right after, the pronunciation of a consonant whose primary point of articulation is not labial.

Palatalised (k^j g^j)

A palatalised consonant is one whose articulation involves the tongue moving towards the roof of the mouth.

Pharyngealised (t^ʕ k^ʕ)

This diacritic symbolises that the sound involves constriction of the pharynx.

Velarised (ɫ ɰ mɰ)

This diacritic means the tongue is raised towards the velum (soft palate) during the production of the sound.

Velarised or Pharyngealised (ɫ ɰ ɾ ɹ)

See the two above. This one is just a bit more versatile.

NOTE

The above diacritics, from *Aspirated* to here, are not written on sounds already implicitly involving those means of articulation. You do not need to make it clearer that a palatal consonant is palatalised, or that a bilabial sound involves the lips.

Raised/Lowered (ɛ ɛ̃)

These diacritics denote a difference in the height of the tongue, compared to a given point of reference, when producing a sound.

Advanced/Retracted Tongue Root (ɛ ɛ̃)

This diacritic means that the tongue's root is moved forward (advanced) or backwards (retracted) for the production of a sound.

This has the consequence of making the pharynx more open (advanced tongue root) or more closed (retracted tongue root).

Dental (ɬ ɳ)

This diacritic is used to mean that the tongue's tip or blade is touching the upper teeth.

Apical (ɬ ɬ̟)

Marks a sound produced by blocking the air flow with the *tip* of the tongue.

Laminal (ɬ ɬ̟)

Marks a sound produced by blocking the airflow with the *blade* of the tongue.

Nasalised (ẽ ẽ̃)

Marks a sound produced by diverting part of the airflow towards the nasal cavity.

Nasal release (dⁿ tⁿ)

Indicates that a plosive consonant is not fully released, instead directing the airflow through the nose.

Lateral release (t¹ ɟ¹)

Indicates that a plosive consonant is released laterally, meaning that the airflow is released, after the occlusion (closure of the vocal tract), on the sides of the tongue while the tip or blade remains in contact with the secondary articulator.

No audible release (k̚ t̚)

Only applied to stop consonants (plosives and nasals), this diacritic marks that sound production stops at the occlusion.

Double articulation and Affricates (k͡p t͡ʃ)

The tie bar is used to unify two symbols as belonging to a single unit of sound.

Suprasegmentals

Suprasegmentals indicate not how each individual sound is pronounced, but how they behave as groups: tone, stress, intonation, length... Those groups of sounds can be syllables, or entire words or phrases. They are called "suprasegmentals" because they are properties of units larger than *the segment*, which is the minimal unit.

The names of each diacritic in this group should be explicit enough that they don't require further explanation... For now.

For the vowel length diacritics, a normal vowel (e) is "short" and has a length of 1, while a half-long vowel (e ·) has a length of 1.5, a long vowel (e :) has a length of 2, and an extra-short vowel (ẽ) has a length of 0.5.

Those numbers aren't fixed and can change between languages and context, but they give a general idea of the relation between the lengths.

Intonation & tones

Tone is a distinctive pitch added to a syllable. It can be phonemically relevant, sometimes conveying grammatical or lexical information.

Tone can happen in languages to contrast sounds. For instance, a vowel with a higher tone would be understood differently in a word from a vowel with a lower tone.

The global rise and global fall arrows indicate a change in the overall pitch of the word, phrase or sentence. For instance, a global rise arrow could be put at the end of most transcriptions of an English interrogative sentence.

Typing the IPA

You can use multiple tools to help you type the IPA, as its symbols are rarely readily available on standard keyboards.

- [List of unicode codes of IPA symbols](#)
- [Clickable IPA chart & virtual keyboard](#)
- [IPA TypeIt, virtual keyboard](#)
- [IPA character picker](#)
- [Another character picker](#)
- [A keyboard layout for Windows](#)
- [Several keyboard layouts for Windows, Linux and MacOS](#)
- [GBoard, a mobile keyboard](#)

Having fun

Having fun is a great way to learn! Open an IPA chart in one tab and [the Pink Trombone](#) in the other, and figure it out!

Getting into it

For ease of use, all of the following will be using a General American English pronunciation, as it is a widespread dialect that is heavily represented in media and everyone is likely to have heard it.

Notation

Throughout the whole course, we are going to be using various bracket types to enclose words. Whether those words are written using the language's writing system, a [romanisation](#), phonemes or phones.

Orthography & romanisation

The native orthography of a language is the one used by its speakers to represent it in writing. It can be different from a *romanisation*, which is the use of the Latin alphabet to represent it in a way that is more approachable to people who do not know the native system.

Equally, there can be adaptations to other writing systems, such as [cyrillisation](#) or [gairaigo](#).

Generally, both of those are encompassed in angled brackets $\langle \rangle$, but you might want to use the native orthography in those brackets and use regular quotation marks for the romanisation. Or the reverse, I'm not your mum. Just make it clear!

Phonemes & phones

[Phonemes](#), enclosed in forward slashes $//$ are probably the hardest thing to explain or grasp in this document. They are **abstract, meaning-distinguishing units of sound**, and each stands for a **group of possible realisations**.

They are the smallest meaning distinguishing unit within a word. This means changing a *phoneme* in a word can give it another **meaning** that is **distinct**.

Let us consider for instance the *phonemic* word for "tree", written using the IPA: $/tʃi:/$.

You might recognise the symbols used here as:

1. alveolar stop $/t/$ $\langle t \rangle$
2. alveolar approximant $/ʃ/$ $\langle r \rangle$
3. long, closed unrounded front vowel $/i:/$ $\langle ee \rangle$

If you try pronouncing this word as those individual sounds, it probably is fairly close to how you pronounce "tree". You may even be correcting it without thinking about it to correspond exactly to how you say the word "tree".

However, we do **not** pronounce *phonemes*: we pronounce [phones](#).

Let's now delve a tiny bit further into this, using a common phenomenon across American dialects. For such dialects, we can notate the word "tree" as [tʃɹi:] .

We can note three things:

1. the brackets changed
2. ʃ was added
3. a diacritic notating retraction was added under the ɹ

The brackets changed because, while this is how an American English speaker would very likely pronounce the word "tree", not all of the information between those brackets is *phonemic*. Instead, we will say it is *phonetic*. As such, the symbols stand for *sounds* and not *phonemes*: **those are actually pronounced**.

We could have rendered the <r> sound more complexly than [ɹ] , as it is **a very complicated sound and can involve a lot of detail**. For some speakers, it could be notated [v] while for others [ɹ̠^w] could be used.

The fact that *phonemes* are pronounced differently depending on their *phonological context* (i.e. which sounds surround them) is called *allophony*.

It can be said that, in the dialect used above, [tʃ] **is an allophone of /t/** that occurs before /ɹ/.

Other easily noticed examples of allophony include /h/ shifting to the palatal fricative [ç] before /ju/, such as in <hue> /hju:/ [çu:]; or /t/ shifting to the alveolar tap [ɾ] between a stressed vowel and an unstressed one, for instance with <butter> /ˈbʌtəɪ/ [ˈbʌɾə].

Phonemes can undergo changes. Those changes can depend on the *phonological context* — which phonemes are around the one(s) being changed, as well as syllable or word boundaries. While *phonemes* are represented in slashes // , actual pronunciation (*phones*) is notated between square brackets [] .

Here is one last example to display all three notations: <conlang> /kanlæŋ/ [kʰanl̠eəŋ] .

Here, the /k/ has become *aspirated* [kʰ], the /l/ is *velarised* into [l̠] and **Allen's horrible accent** has him pronounce /æ/ as [eə] before a nasal. We hope he gets better soon.



NOTE

You probably will see the two methods of transcription above referred to as "broad" (//) and "narrow" ([]). While those are accepted terms in linguistics, I find that they are very unhelpful to *explain* the differences between /phonemes/ and [phones], because a [phonetic transcription] can vary a lot in its level of detail, which can be said to be *narrowness*. See for instance the details that can be involved in the transcription of the r-sound, described a bit above.

Identifying phonemes

While this part is not very relevant to conlanging because it has more to do with linguistic analysis and conlanging rarely can take that route as there is no speakers or corpus to analyse, it's still good to know how this works. And nothing is stopping you from creating a corpus to analyse!

Functional analysis

Let us consider the name "Paul" and that it is pronounced [pɔːl]. This should be reminiscent of the above example that used "conlang" in that the [l] sound is velarised. Now, let's change one of these sounds! You should be able to guess the English words behind those two examples:

- [pɔːn]
- [pɔːz]

If you had "paw" and "pause" or "paws", you were right! As this singular change in the sounds of the word has an effect on the meaning, despite each sound being in the same place in the word, it can be said that those sounds are *distinctive*. This distinction in turn means that those sounds constitute different *phonemes*. Such words that differ only by one sound but have different meanings are called **minimal pairs**.

Distributional analysis

Let us now consider the following sentence: "Liz claims that Paul lied about the play". Let us also consider its phonetic transcription: [lɪz klɛɪmz ðæt pɔːl lɪəd əbaʊt ðə pleɪ].

We look closer at the variations of l that occur, and we can count three: [l], [ɫ] and [ɭ].

By yourself, before looking at the rest of this page, try to identify the contexts in which each occurs.

- [l] occurs before vowels
- [ɫ] occurs after p and k
- [ɭ] occurs after vowels

These sounds occur in describable and predictable contexts, which are complementary: where one of these sounds occurs, we would not expect to hear (or see in transcription) another one of them in its stead. This is called *complementary distribution*. If you tried swapping them around and apply them in contexts where they don't belong, the words would simply sound weird, but none of the meaning would be lost.

As those sounds are *phonetically similar* and occur in *complementary distribution*, they must belong to a same group. This group is the *phoneme*.

The question now becomes: how do we represent the phoneme, which will stand for all of those sounds?

In order to select an adequate representation of this abstract unit, we generally look at the most simple character available in the roman alphabet, which the IPA is based on. Here, the phoneme would be /l/.

We can now re-write our starting sentence using a *phonemic transcription*: /lɪz kleɪms ðæt pɔːl laɪd əbaʊt ðə pleɪ/. But now, we know that each of those /l/ stands for a variety of (*allo*)*phones*, and that **those realisations are dependent on the sounds that surround them**.

NOTE

Remember when I said that the [phonetic transcriptions] could be as broad or narrow as one wanted them to be and that I disliked "broad" and "narrow" as terms describing respectively /phonemic/ and [phonetic] transcriptions?

Well, here's why: in all the examples of phonetic transcription of this chapter so far, I've omitted, among other things, to notate aspiration for the plosives. It could have been [p^hɔːt], [p^hɔːk] and [p^hɔːz], but I chose not to include that much phonetic detail, thus making my notation fairly broad despite being categorised as "narrow transcription".

Minimal pairs

Let's take the concept of minimal pairs, explained a bit above, even further! Again, with the word "Paul" [p^hɔːt] (with aspiration marked this time, because we'll use it!).

In order to know whether or not a given sound is important for the meaning of the word, and thus phonemic, we can proceed by comparing the word with other words that are phonetically similar. For this one, it's better to make sure you can pronounce both [p] and [p^h].

We'll change the sounds one by one.

1	2	3
[k ^h ɔːt]	[p ^h ɪt]	[p ^h ɔːz]
[t ^h ɔːt]	[p ^h eɪt]	[p ^h ɔːn]
[mɔːt]	[p ^h ɜːt]	[p ^h ɔːt]
[pɔːt]	[p ^h ɔːt]	[p ^h ɔːl]

The three top rows yield phonetic words that we can understand as different from "Paul", divided here by column:

1. call, tall, mall;
2. pill, pale, pearl;
3. pause/paws, pawn, port.

The last word, "port" is cheating a little: only dialects that do not pronounce the *rhotic* (r-sound) in the word, such as British English, would realise it this way.

The final row, however, does not contain identifiable words that are different from "Paul". They're "Paul, but weird". We can identify the differences:

1. without aspiration
2. raising of the vowel
3. the allophone of /l/ here is not correct for its context (see the previous section, "Distributional analysis").